

$$\{e\} \triangleleft A_5 \triangleleft S_5$$

pero A_5 no es cíclico
(abeliano)

A_5 es un grupo
simple

$L:K$ "soluble"

$L \subset \underbrace{(F)}_{\text{radical}}$

$$K = K_0 \subset K_1 \subset \dots \subset K_n = F$$

Abel-Galois

Si $L:K$ es soluble

$\Rightarrow \text{Gal}(L/K)$ es soluble

$$K_i = K_{i-1}(\sqrt[n_i]{a})$$

$$a \in K_{i-1}$$

Teorema

Sea $L : K$ Galois tal que $\text{Gal}(L/K) \simeq \mathbb{Z}/p\mathbb{Z}$. Si K contiene una raíz p -ésima de la unidad, entonces existe $\alpha \in L$ tal que $L = K(\alpha)$ y $\alpha^p \in K$

$$(\mathbb{Z}/p\mathbb{Z}, +) = \langle 1 \rangle$$

Dem: (Cox)

$$\rho_w \text{ Gal}(L/K) \simeq \mathbb{Z}/p\mathbb{Z}$$

$$K \subset L$$

$$L = K(\sqrt[p]{a})$$

$$= K(\alpha)$$

$$\alpha^p \in K (a \in K)$$

existe $\sigma \in \text{Gal}(L/K)$; $\langle \sigma \rangle = \text{Gal}(L/K)$

$$\text{así } \sigma^p = e \quad \left| \sigma: L \rightarrow L, \sigma^p = \text{id}_L \right|$$

Consider a σ como un K -transf. Lineal

$$\begin{array}{l} L \\ | \\ K \end{array} \quad \begin{array}{l} \sigma(a+b) = \sigma(a) + \sigma(b) \\ \sigma(ka) = k\sigma(a) \end{array} \quad \checkmark$$

Assi σ , una trans. lineal, satisfice
el pol $X^p = 1$ ($\sigma^p = 1_L$)

$X^p - 1 = 0$. σ tiene p valores

propios $1, \zeta, \zeta^2, \dots, \zeta^{p-1}$ ($\zeta^p = 1$)

y así tiene vectores propios

$\exists \alpha \in L$; $\sigma(\alpha) = \zeta \alpha$. ($\alpha \in K$?)

$\alpha \in L \setminus K$ $\sigma(\alpha^p) = \sigma(\alpha)^p$
 $= \zeta^p \alpha^p = \alpha^p$

$\sigma(\alpha^p) = \alpha^p$

Por lo tanto vector
propio $\alpha^p \in K$

$$\text{Assi } \alpha \in L \setminus K \quad \alpha^p = \beta \in K$$

$$\alpha = \sqrt[p]{\beta} \quad \text{y} \quad L = K(\alpha) = K(\sqrt[p]{\beta})$$

$$[L:K] = |\text{Gal}(L/K)| = p$$

$$\underbrace{K \subset K(\alpha)}_{\neq 1} \subset \underbrace{L}_{\neq 1} \quad \text{Assi } \underline{L = K(\alpha)}$$

$$L = K(\sqrt[p]{\beta})$$

$$\beta = \alpha^p \in K$$

□

Resolver $X^2 + bX + c = 0$

Usando Galois $X^2 + bX + c = (X - \alpha)(X - \beta)$

$$\alpha + \beta = -b \quad \alpha\beta = c.$$

asi el cuerpo de descomp es $\mathbb{Q}(\alpha)$

$$\mathbb{Q} \xrightarrow{2} \mathbb{Q}(\alpha) \mid \mathbb{Q}(\alpha): \mathbb{Q} \text{ es Galois}$$

Ahi se encuentra un vector propio de
un elemento de $\text{Gal}(\mathbb{Q}(\alpha)/\mathbb{Q})$ (vch. prop $\neq 1$)

$$\sigma \in \text{Gal}(\mathbb{Q}(\alpha)/\mathbb{Q}) \quad |\text{Gal}(\mathbb{Q}(\alpha)/\mathbb{Q})| = 2$$

$$\sigma \neq \text{id} \quad \sigma(\alpha) = \alpha, \beta \quad \sigma(\alpha) = \beta$$

$$\sigma(\alpha - \beta) = \beta - \alpha = -(\alpha - \beta)$$

$$\boxed{\sigma(\alpha - \beta) = (-1)(\alpha - \beta)} \quad \left\{ \begin{array}{l} x^2 + bx + c = 0 \end{array} \right.$$

$$(\alpha - \beta)^2 \in \mathbb{Q}$$

$$(\alpha - \beta)^2 = \alpha^2 - 2\alpha\beta + \beta^2$$

$$\boxed{(\alpha - \beta)^2 = (-b)^2 - 4c}$$

$$= (\alpha + \beta)^2 - 4\alpha\beta$$

$$= (-b)^2 - 4c$$

$$\mathbb{Q}(\alpha) = \mathbb{Q}(\sqrt{b^2 - 4c})$$

$$\left. \begin{array}{l} \alpha - \beta = \sqrt{b^2 - 4c} \\ \alpha + \beta = -b \end{array} \right\} \alpha = \frac{-b + \sqrt{b^2 - 4c}}{2}$$

Objeto: $X^3 + bx + c = 0$ (irreducible)
y sea L el cuerpo de descomp.

$$[L:\mathbb{Q}] = 3, 6. \quad \boxed{\text{Gal}(L/\mathbb{Q}) \cong S_3, A_3}$$

$$\text{Supongo } \text{Gal}(L/\mathbb{Q}) = A_3 = \langle (123) \rangle \\ = (\mathbb{Z}/3\mathbb{Z}, +)$$

$$\text{Así } \exists \sigma \in \text{Gal}(L/\mathbb{Q}) = \langle \sigma \rangle$$

y todo u en L tiene que haber u en L en donde
un vector propio con valor propio $\neq 1$

$$L/\mathbb{Q} \quad \underbrace{L/\mathbb{Q}(w)}_{\substack{3 \\ \text{grado}}} \quad w^2 + w + 1 = 0.$$

$$X^3 + bx + c = (X - \alpha)(X - \beta)(X - \gamma)$$

$$L = \mathbb{Q}(\omega)(\alpha, \beta, \gamma)$$

$$V = \alpha + \beta + \gamma = 0.$$

$$\sigma(V) = V.$$

Lagrange

$$\left\{ \begin{array}{l} \sigma(V) = \omega V \\ \Downarrow \\ V^3 \in \mathbb{Q}(\omega) \end{array} \right. \quad \left\{ \begin{array}{l} \sigma(\alpha) = \beta \\ \sigma(\beta) = \gamma \\ \sigma(\gamma) = \alpha \end{array} \right.$$

$$V_1 = \alpha + \omega \beta + \omega^2 \gamma.$$

$$\sigma(V_1) = \beta + \omega \gamma + \omega^2 \alpha = \omega^2 (\alpha + \omega \beta + \omega^2 \gamma) = \omega^2 V_1$$

$$V_1^3 \in \mathbb{Q}(\omega)$$

$$V_2 = \alpha + \omega^2 \beta + \omega \gamma$$

$$\sigma(V_2) = \beta + \omega^2 \gamma + \omega \alpha = \omega (\alpha + \omega^2 \beta + \omega \gamma)$$

$$\alpha + \beta + \gamma = 0$$

$$(\alpha + \omega\beta + \omega^2\gamma)^3 \in \mathbb{Q}(\omega)$$

$$(\alpha + \omega^2\beta + \omega\gamma)^3 \in \mathbb{Q}(\omega)$$

$$\begin{cases} \alpha + \beta + \gamma = 0 \\ \alpha + \omega\beta + \omega^2\gamma = \sqrt[3]{A} \\ \alpha + \omega^2\beta + \omega\gamma = \sqrt[3]{B} \end{cases} \quad A, B \in \mathbb{Q}(\omega)$$

$$3\alpha = \sqrt[3]{A} + \sqrt[3]{B} \quad \alpha = \frac{\sqrt[3]{A} + \sqrt[3]{B}}{3}$$

$$\text{Gal}(L/K) = S_3$$

$$\{e\} \leq A_3 \leq S_3 \quad A_3 \trianglelefteq S_3$$

$$A_3 = (\mathbb{Z}/3\mathbb{Z}, +) \quad \gamma \quad S_3/A_3 = (\mathbb{Z}/2\mathbb{Z}, +)$$

$$\mathbb{Q}(\omega, \alpha, \beta, \gamma) \supseteq \underline{\mathbb{Q}} \supseteq \mathbb{Q}(\omega)$$

$$\text{Gal}(\mathbb{Q}/\mathbb{Q}(\omega)) = (\mathbb{Z}/2\mathbb{Z}, +)$$

Extensiones Ciclotómicas

$$\mathbb{Q}(\zeta_n)$$

Polinomio Ciclotómico

Sea $\zeta_n = e^{2\pi i/n}$, entonces tenemos que:

$$x^n - 1 = \prod_{0 \leq i < n} (x - \zeta_n^i).$$

Definimos el *polinomio ciclotómico* n -ésimo $\Phi_n(x)$ como el producto:

$$\Phi_n(x) = \prod_{\substack{0 \leq i < n \\ \text{mcd}(i, n) = 1}} (x - \zeta_n^i)$$

Así, las raíces de $\Phi_n(x)$ son los ζ_n^i tales que $0 \leq i < n$ y $\text{mcd}(i, n) = 1$.

Por lo tanto $\Phi_n(x)$ tiene grado $\varphi(n)$, donde φ es la función de Euler.

$$\varphi(n) = \deg(\Phi_n(x))$$

$$\Phi_1(x) = x - 1$$

$$\Phi_2(x) = x - (-1)^1 = x + 1 \quad \zeta_2 = -1 = e^{\frac{2\pi i}{2}}$$

$$\begin{aligned} \Phi_3(x) &= (x - \omega)(x - \omega^2) \\ &= x^2 + x + 1 \end{aligned} \quad \begin{aligned} \omega^3 &= 1 \\ \omega &\neq 1 \end{aligned}$$

$$\Phi_4(x) = (x-i)(x-i^3) \quad \zeta_4 = e^{2\pi i/4} = i$$

$$= x^2 + 1$$

$$\Phi_5(x) = x^4 + x^3 + x^2 + x + 1$$

$$\Phi_6(x) = \frac{x^6 - 1}{(x^2 - 1)(x^2 + x + 1)}$$

$$\frac{x^3 - 1}{x - 1}$$

$$= \frac{x^4 + x^2 + 1}{x^2 + x + 1}$$